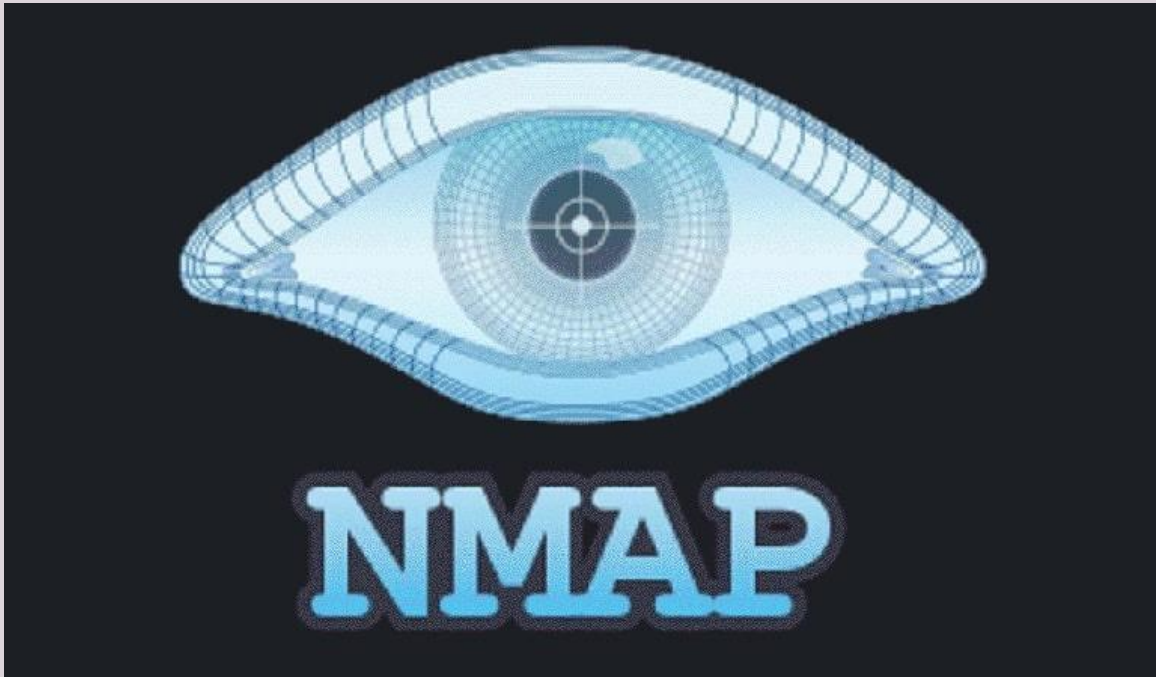




NMAP WALKTHROUGH

NETWORK EXPLORATION TOOL

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Installation

- Nmap was originally designed for Unix/Linux systems, but now it also works on Windows and macOS also.
- Most popular Linux distributions provide binary Nmap packages which allow for simple installation.

Syntax - Command: apt install nmap

```
(root@abhishek)-[/home/kali]
# apt install nmap
Upgrading:
nmap nmap-common

Summary:
  Upgrading: 2, Installing: 0, Removing: 0, Not Upgrading: 1963
  Download size: 6269 kB
  Space needed: 287 kB / 84.6 GB available

Continue? [Y/n] y
Get:1 http://http.kali.org/kali kali-rolling/non-free amd64 nmap amd64 7.94+git20230807.3be01efb1+dfsg-4kali3 [1936 kB]
Get:2 http://http.kali.org/kali kali-rolling/non-free amd64 nmap-common all 7.94+git20230807.3be01efb1+dfsg-4kali3 [4333 kB]
Fetched 6269 kB in 4s (1492 kB/s)
(Reading database ... 390929 files and directories currently installed.)
Preparing to unpack .../nmap_7.94+git20230807.3be01efb1+dfsg-4kali3_amd64.deb ...
Unpacking nmap (7.94+git20230807.3be01efb1+dfsg-4kali3) over (7.94+git20230807.3be01efb1+dfsg-2+kali2+b1) ...
Preparing to unpack .../nmap-common_7.94+git20230807.3be01efb1+dfsg-4kali3_all.deb ...
Unpacking nmap-common (7.94+git20230807.3be01efb1+dfsg-4kali3) over (7.94+git20230807.3be01efb1+dfsg-2+kali2) ...
Setting up nmap-common (7.94+git20230807.3be01efb1+dfsg-4kali3) ...
Setting up nmap (7.94+git20230807.3be01efb1+dfsg-4kali3) ...
Setcap worked! Adding configuration to environment
Processing triggers for kali-menu (2023.4.7) ...
Processing triggers for man-db (2.12.1-1) ...
Processing triggers for wordlists (2023.2.0) ...
```

Note: If it's already installed, you can either choose to upgrade it or type 'no' to stop the installation.

- Once Nmap has been successfully installed, you can verify it is working correctly by executing **#nmap localhost** on the command line.

```
(root@abhishek)-[/home/kali]
# nmap localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-11-14 15:58 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000070s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds
```

Basic Scanning Overview

This section covers the basics of network scanning with Nmap. Before we begin it is important to understand the following concepts:

- Firewalls, routers, proxy servers, and other security devices can skew the results of an Nmap scan.
- Some scanning options require elevated privileges. On Unix and Linux systems you may be required to login as the root user or to execute Nmap using the sudo command.

➤ Scan Single Target

- A target can be specified as an IP address or host name which Nmap will try to resolve.

Syntax: #nmap [target]

E.g.: #nmap 192.168.1.1

```
(root@abhishek)-[/home/kali]
# nmap 192.168.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-11-14 16:06 IST
Nmap scan report for 192.168.1.1
Host is up (0.0037s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 17.59 seconds
```

➤ Scan Multiple Targets

Nmap can be used to scan multiple hosts at the same time. The easiest way to do this is to string together the target IP addresses or host names on the command line (separated by a space).

Syntax -#nmap [target1 target2 etc]

Eg. #nmap 192.168.10.1 192.168.10.100 192.168.10.101

```

(root@abhishek)-[/home/kali]
# nmap 192.168.10.1 192.168.10.100 192.168.10.101
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-11-14 16:09 IST
Nmap scan report for 192.168.10.1
Host is up (0.0043s latency).
All 1000 scanned ports on 192.168.10.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap scan report for 192.168.10.100
Host is up (0.0039s latency).
All 1000 scanned ports on 192.168.10.100 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap scan report for 192.168.10.101
Host is up (0.0039s latency).
All 1000 scanned ports on 192.168.10.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 3 IP addresses (3 hosts up) scanned in 41.00 seconds

```

➤ Scan a Range of IP Addresses

A range of IP addresses can be used for target specification as demonstrated in the example below.

Syntax: #nmap [Range of IP addresses]

E.g.: #nmap 192.168.1-100

➤ Scan a List of Targets

If you have a large number of systems to scan, you can enter the IP address (or host names) in a text file and use that file as input for Nmap on the command line.

```

(root@abhishek)-[/home/kali]
# nano list.txt

(root@abhishek)-[/home/kali]
# cat list.txt
192.168.1.1
192.168.1.2
192.168.1.3
192.168.1.4
192.168.1.5
192.168.1.6
192.168.1.7
192.168.1.8

```

The list.txt file above contains a list of hosts to be scanned. Each entry in the list.txt file must be separated by a space, tab, or newline. The `-iL` parameter is used to instruct Nmap to extract the list of targets from the list.txt file.

```
[sudo] password for kali:
(root@abhishek)-[/home/kali]
# nmap -iL list.txt
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-11-14 16:24 IST
Nmap scan report for 192.168.1.1
Host is up (0.15s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap scan report for 192.168.1.2
Host is up (0.029s latency).
All 1000 scanned ports on 192.168.1.2 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap scan report for 192.168.1.3
Host is up (0.070s latency).
All 1000 scanned ports on 192.168.1.3 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap scan report for 192.168.1.4
Host is up (0.052s latency).
All 1000 scanned ports on 192.168.1.4 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap scan report for 192.168.1.5
Host is up (0.16s latency).
All 1000 scanned ports on 192.168.1.5 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap scan report for 192.168.1.6
Host is up (0.14s latency).
All 1000 scanned ports on 192.168.1.6 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap scan report for 192.168.1.7
Host is up (0.062s latency).
All 1000 scanned ports on 192.168.1.7 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap scan report for 192.168.1.8
Host is up (0.068s latency).
All 1000 scanned ports on 192.168.1.8 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 8 IP addresses (8 hosts up) scanned in 346.59 seconds
```

➤ Exclude Targets from a Scan

The `--exclude` option is used with Nmap to exclude hosts from a scan.

syntax: `nmap [targets] --exclude [target(s)]`

Eg. `Nmap 192.168.1.1/24 --exclude 192.168.1.10`

Note: The `--exclude` option is useful if you want to exclude specific hosts when scanning a large number of addresses. In the example above host 192.168.10.100 is excluded from the range of targets being scanned.

➤ Perform an Aggressive Scan

The `-A` parameter instructs Nmap to perform an aggressive scan.

Syntax: `#nmap -A [target]`

E.g.: `#nmap -A 192.168.1.1`

```
(root@abhishek)-[/home/kali]
# nmap -A 192.168.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-11-14 16:38 IST
Nmap scan report for 192.168.1.1
Host is up (0.014s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: 3Com 4500G switch (92%), H3C Comware 5.20 (92%), Huawei VRP 8.100 (92%), Microsoft Windows Server 2003 SP1
way (92%), AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 2 hops

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 2.19 ms 10.0.2.2
2 2.19 ms 192.168.1.1
```

➤ Scan an IPv6 Target

The `-6` parameter is used to perform a scan of an IP version 6 target.

Syntax: `#nmap -6 [target]`

E.g.: `#nmap -6 fe80::29aa:9db9:4164:d80e`

```
(root@abhishek)-[/home/kali]
# nmap -6 fe80::29aa:9db9:4164:d80e
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-11-14 16:43 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.56 seconds
```

Basic Scanning Functions Commands use in Nmap:

Feature	Option
Don't Ping	-PN
Perform a Ping Only Scan	-sP
TCP SYN Ping	-PS
TCP ACK Ping	-PA
UDP Ping	-PU
SCTP INIT Ping	-PY
ICMP Echo Ping	-PE
ICMP Timestamp Ping	-PP
ICMP Address Mask Ping	-PM
IP Protocol Ping	-PO
ARP Ping	-PR
Traceroute	--traceroute
Force Reverse DNS Resolution	-R
Disable Reverse DNS Resolution	-n
Alternative DNS Lookup	--system-dns
Manually Specify DNS Server(s)	--dns-servers
Create a Host List	-sL

Advance Scanning Functions Commands use in Nmap:

Feature	Option
TCP SYN Scan	-sS
TCP Connect Scan	-sT
UDP Scan	-sU
TCP NULL Scan	-sN
TCP FIN Scan	-sF
Xmas Scan	-sX
TCP ACK Scan	-sA
Custom TCP Scan	--scanflags
IP Protocol Scan	-sO
Send Raw Ethernet Packets	--send-eth
Send IP Packets	--send-ip

Note: You must login with root/administrator privileges (or use the sudo command) to execute many of the scans discussed in this section.

Port Scanning Options

Feature	Option
Perform a Fast Scan	-F
Scan Specific Ports	-p [port]

Scan Ports by Name	-p [name]
Scan Ports by Protocol	-p U:[UDP ports],T:[TCP ports]
Scan All Ports	-p "*"
Scan Top Ports	--top-ports [number]
Perform a Sequential Port Scan	-r

Operating System and Service Detection

Feature	Option
Operating System Detection	-O
Attempt to Guess an Unknown OS	--osscan-guess
Service Version Detection	-sV
Perform a RPC Scan	--version-trace
Troubleshooting Version Scans	-sR

Timing Options

Feature	Option
Timing Templates	-T[0-5]
Set the Packet TTL	--ttl
Minimum # of Parallel Operations	--min-parallelism
Maximum # of Parallel Operations	--max-parallelism
Minimum Host Group Size	--min-hostgroup
Maximum Host Group Size	--max-hostgroup
Maximum RTT Timeout	--max-rtt-timeout
Initial RTT Timeout	--initial-rtt-timeout
Maximum Retries	--max-retries
Host Timeout	--host-timeout
Minimum Scan Delay	--scan-delay
Maximum Scan Delay	--max-scan-delay
Minimum Packet Rate	--min-rate
Maximum Packet Rate	--max-rate
Defeat Reset Rate Limits	--defeat-rst-ratelimit

➤ Timing Templates

The -T parameter is used to specify a timing template for an Nmap scan.

Usage syntax: `nmap -T[0-5] [target]`

E.g.: `#nmap -t4 192.168.1.1`

Template	Name	Notes
-T0	paranoid	Extremely slow
-T1	sneaky	Useful for avoiding intrusion detection systems
-T2	polite	Unlikely to interfere with the target system
-T3	normal	This is the default timing template
-T4	aggressive	Produces faster results on local networks
-T5	insane	Very fast and aggressive scan

Firewall Evasion Techniques Overview

Firewalls and intrusion prevention systems are designed to prevent tools like Nmap from getting an accurate picture of the systems they are protecting. Nmap includes a number of features designed to circumvent these defences. This section discusses the various evasion techniques built into Nmap.

Feature	Option
Fragment Packets	-f
Specify a Specific MTU	--mtu
Use a Decoy	-D
Idle Zombie Scan	-sI
Manually Specify a Source Port	--source-port
Append Random Data	--data-length
Randomize Target Scan Order	--randomize-hosts
Spoof MAC Address	--spoof-mac
Send Bad Checksums	--badsum

Output Options Overview

Feature	Option
Save Output to a Text File	-oN
Save Output to a XML File	-oX
Grepable Output	-oG
Output All Supported File Types	-oA
Periodically Display Statistics	--stats-every
133t Output	-oS

Troubleshooting and Debugging

Technical problems are an inherent part of using computers. Nmap is no exception. Occasionally a scan may not produce the output you expected, you may receive an error – or you may not receive any output at all. Nmap offers

several options for tracing and debugging a scan which can help identify why this happens. The following section describes these troubleshooting and debugging features in detail.

Feature	Option
Getting Help	-h
Display Nmap Version	-V
Verbose Output	-v
Debugging	-d
Display Port State Reason	--reason
Only Display Open Ports	--open
Trace Packets	--packet-trace
Display Host Networking	--iflist
Specify a Network Interface	-e

Conclusion

Nmap is a versatile and powerful network scanning tool, widely used in cybersecurity for discovering hosts, services, and vulnerabilities. Its extensive features and support for scripting make it an essential tool for network administrators and penetration testers. However, while it performs best on Unix/Linux systems, users may encounter minor issues on other platforms.

We can also use the tools like: -

- Angry Ip scanner
- Nessus
- Port scan
- zenmap
- masscan
- Netcat, etc.

Nmap Cheat Sheet

Basic Scanning Techniques	
Scan a Single Target	<code>nmap [target]</code>
Scan Multiple Targets	<code>nmap [target1, target2, etc]</code>
Scan a List of Targets	<code>nmap -iL [list.txt]</code>
Scan a Range of Hosts	<code>nmap [range of ip addresses]</code>
Scan an Entire Subnet	<code>nmap [ip address/cdir]</code>
Scan Random Hosts	<code>nmap -iR [number]</code>
Excluding Targets from a Scan	<code>nmap [targets] --exclude [targets]</code>
Excluding Targets Using a List	<code>nmap [targets] --excludefile [list.txt]</code>
Perform an Aggressive Scan	<code>nmap -A [target]</code>
Scan an IPv6 Target	<code>nmap -6 [target]</code>
Discovery Options	
Perform a Ping Only Scan	<code>nmap -sP [target]</code>
Don't Ping	<code>nmap -PN [target]</code>
TCP SYN Ping	<code>nmap -PS [target]</code>
TCP ACK Ping	<code>nmap -PA [target]</code>
UDP Ping	<code>nmap -PU [target]</code>
SCTP INIT Ping	<code>nmap -PY [target]</code>
ICMP Echo Ping	<code>nmap -PE [target]</code>
ICMP Timestamp Ping	<code>nmap -PP [target]</code>
ICMP Address Mask Ping	<code>nmap -PM [target]</code>
IP Protocol Ping	<code>nmap -PO [target]</code>
ARP Ping	<code>nmap -PR [target]</code>
Traceroute	<code>nmap --traceroute [target]</code>
Force Reverse DNS Resolution	<code>nmap -R [target]</code>
Disable Reverse DNS Resolution	<code>nmap -n [target]</code>
Alternative DNS Lookup	<code>nmap --system-dns [target]</code>
Manually Specify DNS Server(s)	<code>nmap --dns-servers [servers] [target]</code>
Create a Host List	<code>nmap -sL [targets]</code>
Advanced Scanning Functions	
TCP SYN Scan	<code>nmap -sS [target]</code>
TCP Connect Scan	<code>nmap -sT [target]</code>
UDP Scan	<code>nmap -sU [target]</code>
TCP NULL Scan	<code>nmap -sN [target]</code>
TCP FIN Scan	<code>nmap -sF [target]</code>
Xmas Scan	<code>nmap -sX [target]</code>
TCP ACK Scan	<code>nmap -sA [target]</code>
Custom TCP Scan	<code>nmap --scanflags [flags] [target]</code>
IP Protocol Scan	<code>nmap -sO [target]</code>
Send Raw Ethernet Packets	<code>nmap --send-eth [target]</code>
Send IP Packets	<code>nmap --send-ip [target]</code>
Port Scanning Options	
Perform a Fast Scan	<code>nmap -F [target]</code>
Scan Specific Ports	<code>nmap -p [port(s)] [target]</code>
Scan Ports by Name	<code>nmap -p [port name(s)] [target]</code>
Scan Ports by Protocol	<code>nmap -sU -sT -p U:[ports],T:[ports] [target]</code>
Scan All Ports	<code>nmap -p "*" [target]</code>
Scan Top Ports	<code>nmap --top-ports [number] [target]</code>
Perform a Sequential Port Scan	<code>nmap -r [target]</code>
Version Detection	
Operating System Detection	<code>nmap -O [target]</code>
Submit TCP/IP Fingerprints	www.nmap.org/submit/
Attempt to Guess an Unknown	<code>nmap -O --osscan-guess [target]</code>
Service Version Detection	<code>nmap -sV [target]</code>
Troubleshooting Version Scans	<code>nmap -sV --version-trace [target]</code>
Perform a RPC Scan	<code>nmap -sR [target]</code>
Timing Options	

Timing Templates	<code>nmap -T[0-5] [target]</code>
Set the Packet TTL	<code>nmap --ttl [time] [target]</code>
Minimum # of Parallel Operations	<code>nmap --min-parallelism [number] [target]</code>
Maximum # of Parallel Operations	<code>nmap --max-parallelism [number] [target]</code>
Minimum Host Group Size	<code>nmap --min-hostgroup [number] [targets]</code>
Maximum Host Group Size	<code>nmap --max-hostgroup [number] [targets]</code>
Maximum RTT Timeout	<code>nmap --initial-rtt-timeout [time] [target]</code>
Initial RTT Timeout	<code>nmap --max-rtt-timeout [TTL] [target]</code>
Maximum Retries	<code>nmap --max-retries [number] [target]</code>
Host Timeout	<code>nmap --host-timeout [time] [target]</code>
Minimum Scan Delay	<code>nmap --scan-delay [time] [target]</code>
Maximum Scan Delay	<code>nmap --max-scan-delay [time] [target]</code>
Minimum Packet Rate	<code>nmap --min-rate [number] [target]</code>
Maximum Packet Rate	<code>nmap --max-rate [number] [target]</code>
Defeat Reset Rate Limits	<code>nmap --defeat-rst-ratelimit [target]</code>
Firewall Evasion Techniques	
Fragment Packets	<code>nmap -f [target]</code>
Specify a Specific MTU	<code>nmap --mtu [MTU] [target]</code>
Use a Decoy	<code>nmap -D RND:[number] [target]</code>
Idle Zombie Scan	<code>nmap -sI [zombie] [target]</code>
Manually Specify a Source Port	<code>nmap --source-port [port] [target]</code>
Append Random Data	<code>nmap --data-length [size] [target]</code>
Randomize Target Scan Order	<code>nmap --randomize-hosts [target]</code>
Spoof MAC Address	<code>nmap --spoof-mac [MAC 0 vendor] [target]</code>
Send Bad Checksums	<code>nmap --badsum [target]</code>